Political Object Detection

Domain knowledge-guided semi-supervised cluster-then-label approach for political object detection in the wild

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- Framework
  - Task: Political Object Detection
  - Challenges
  - Solutions
  - Resulting data
- Application 1: *Objectscores*
- Application 2: Objects as regression variables

Application 1: *Objectscores*

Liberal Object	Log Odds	Conservative Object	Log Odds
pussy hat	-3.343	orange beanie	3.445
rainbow flag	-2.857	confederate flag	1.712
trans rights now sign	-2.363	american flag	1.400
march for life beanie	-1.870	helmet	1.385
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bright green hat	-1.207	white bike helmet	1.115
pro abortion sign	-0.882	maga hat	1.024
march for lives flag	-0.635	proud boy hat	0.696
black beanie	-0.551	unknown flag	0.425
blue hat	-0.465	american beanie	0.422

Liberal Object	P_{lib}	P_{con}	Score	Conservative Object	P_{lib}	P_{con}	Score
unknown sign	0.247	0.176	-0.071	unknown flag	0.089	0.136	0.047
pussy hat	0.013	0.000	-0.012	american flag	0.011	0.045	0.034
rainbow flag	0.010	0.001	-0.010	helmet	0.003	0.012	0.009
capitol hill building	0.105	0.096	-0.009	maga hat	0.004	0.010	0.007
pro abortion sign	0.008	0.003	-0.005	proud boy hat	0.005	0.011	0.005
police hat	0.004	0.001	-0.003	confederacy flag	0.001	0.006	0.005
bright green hat	0.003	0.001	-0.002	unknown headwear	0.452	0.457	0.004
blue hat	0.005	0.003	-0.002	orange beanie	0.001	0.003	0.003
unknown building	0.010	0.009	-0.001	ski mask	0.001	0.003	0.002
march for lives flag	0.002	0.001	-0.001	cowboy hat	0.004	0.005	0.001

Application 2: Objects as regression variables

	Police Presence
Ideological Agreement	−1.339***
	(0.391)
Num.Obs.	731
R2	
R2 Adj.	
AIC	229.9
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Log.Lik.	−112.974
RMSE	0.19

	(1)	(2)	(3)	(4)
Max Face Size	0.001	−0.002	−0.150+	−0.012
	(0.013)	(0.013)	(0.077)	(0.013)
Crowd Count	0.006***	0.006***	−0.001	0.006***
	(0.001)	(0.001)	(0.006)	(0.001)
Black to Nonblack Ratio	−0.028+	−0.036*	−0.126	−0.029+
	(0.016)	(0.016)	(0.183)	(0.015)
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				(0.011)
Baton Binary				0.008
				(0.015)
Fire Binary				0.194***
				(0.025)

Possible regression variables

- Presence
- Count
- Ratio
- Size
  - Min, Average, Mean, Median, Maximum
- Centrality

What’s a political object?

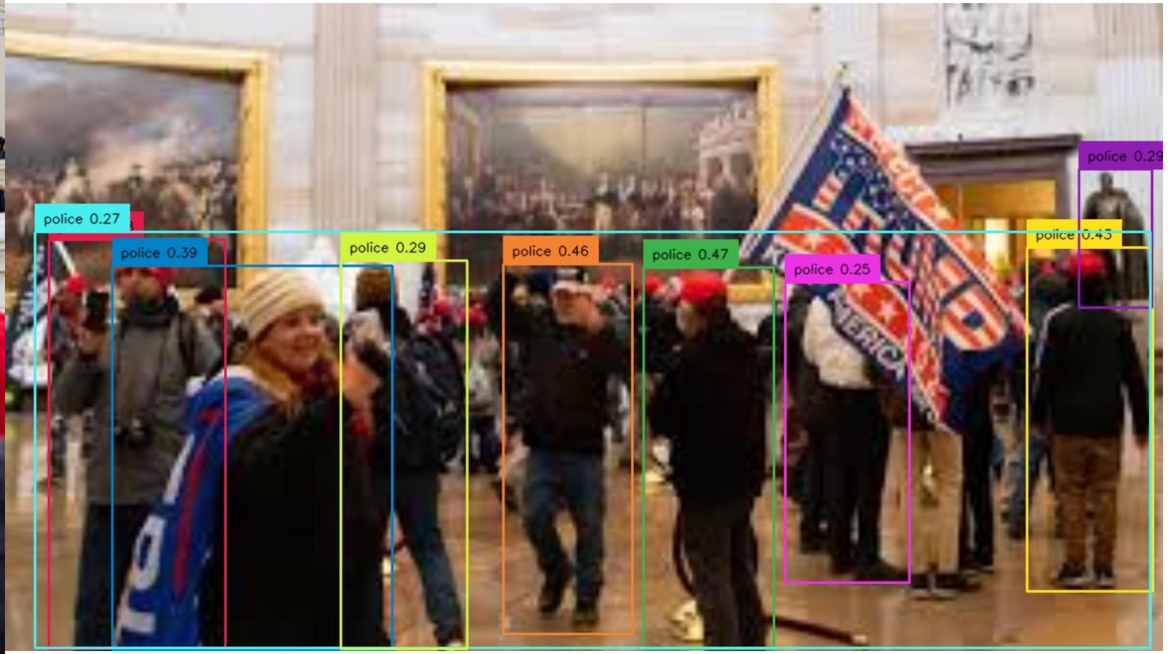
Torres: “protest”
Caprini: {“person”, “flag”, “hat”}
Yun: {“civilian”, “American flag,” “MAGA hat”, “Capitol Hill”}

Torres: “protest”
Caprini: {“person”, “flag”}
Yun: {“civilian”, “Rainbow flag”, “Supreme Court”}

Why detecting political objects matter?

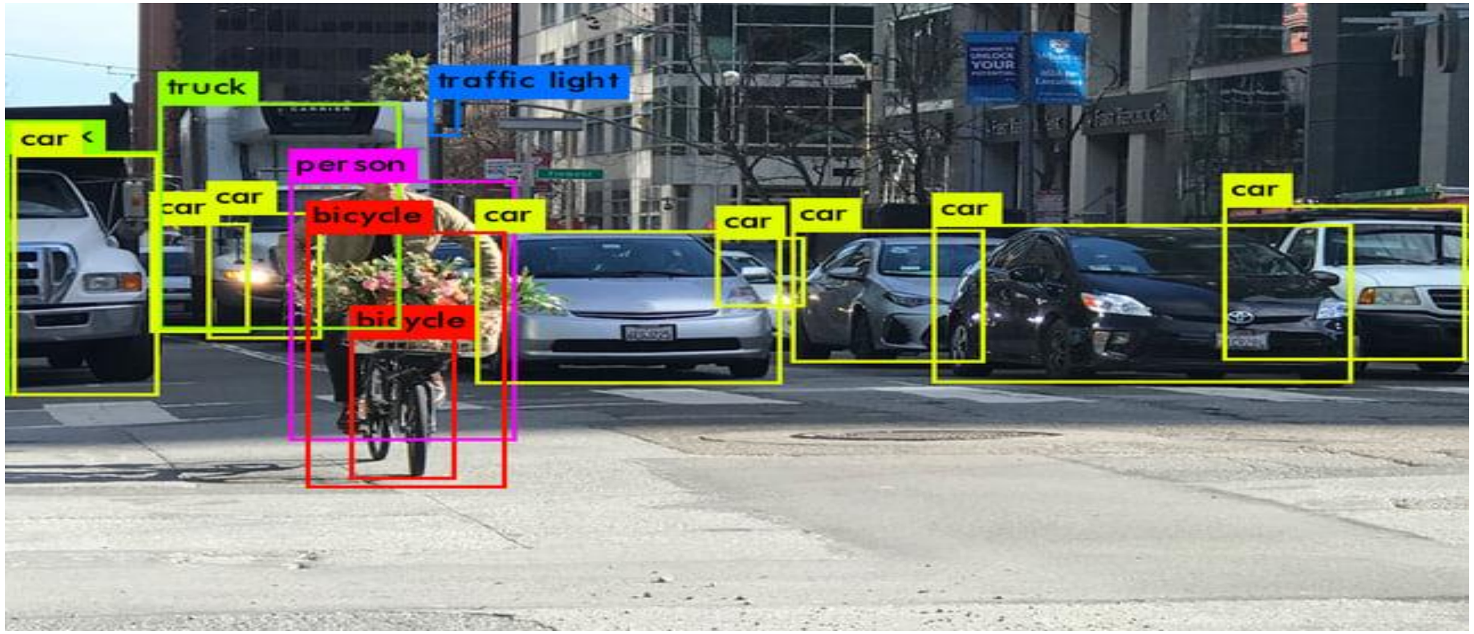
- Scholars of **media**
  - How media outlets portray the same **political event** differently?
  - How media outlets portray the same **issue** differently?
- Scholars of **social movements**
  - How media portrayals of social movements affects public support?
- Scholars of **symbols**
  - How the political meaning of a symbol has changed **over time**?
  - Which symbols are used by **extremists**?

General models can’t detect political objects

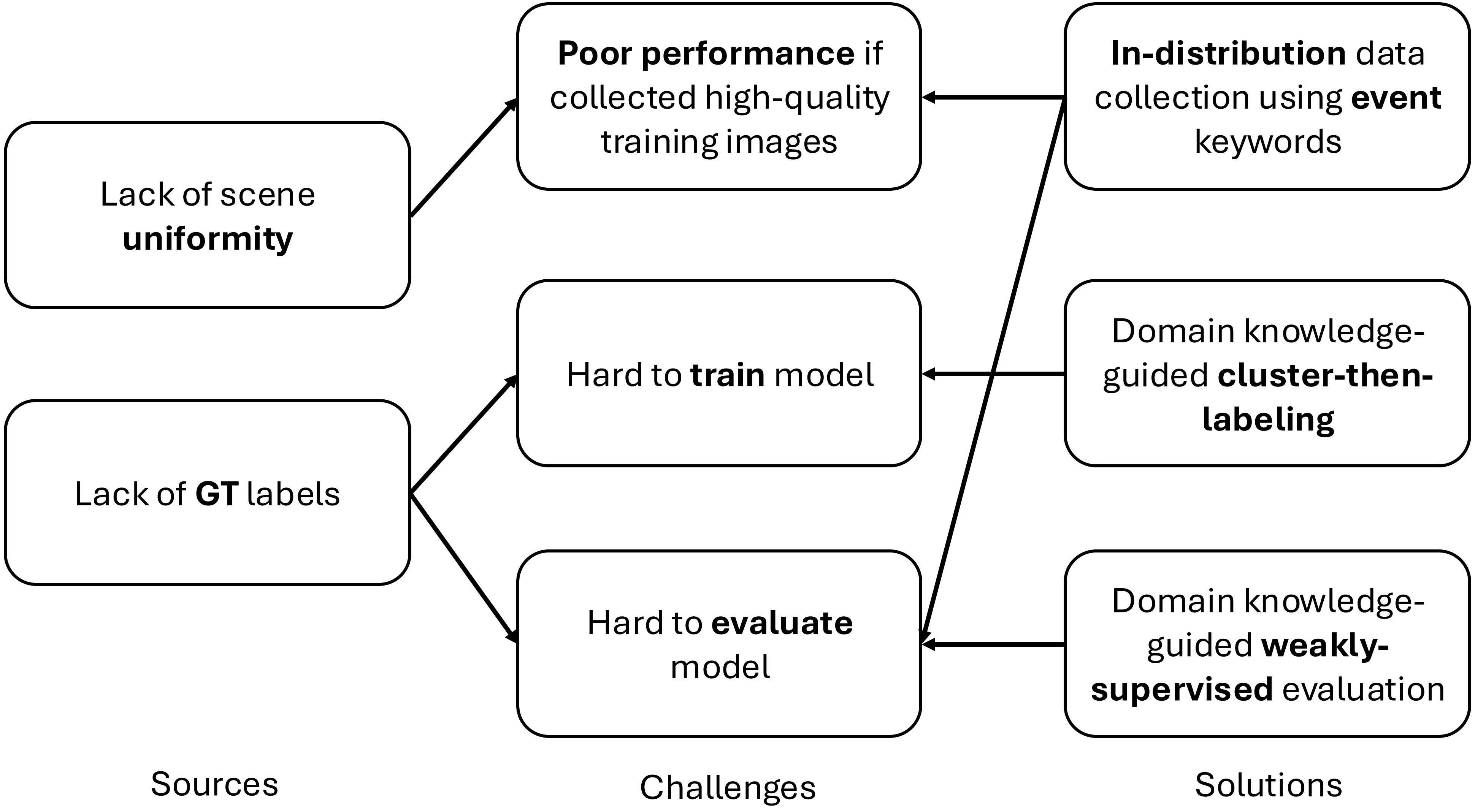


How to develop a customized object detection tool

Data collection → Data annotation → Training (Fine-tuning) → Evaluation → Deployment



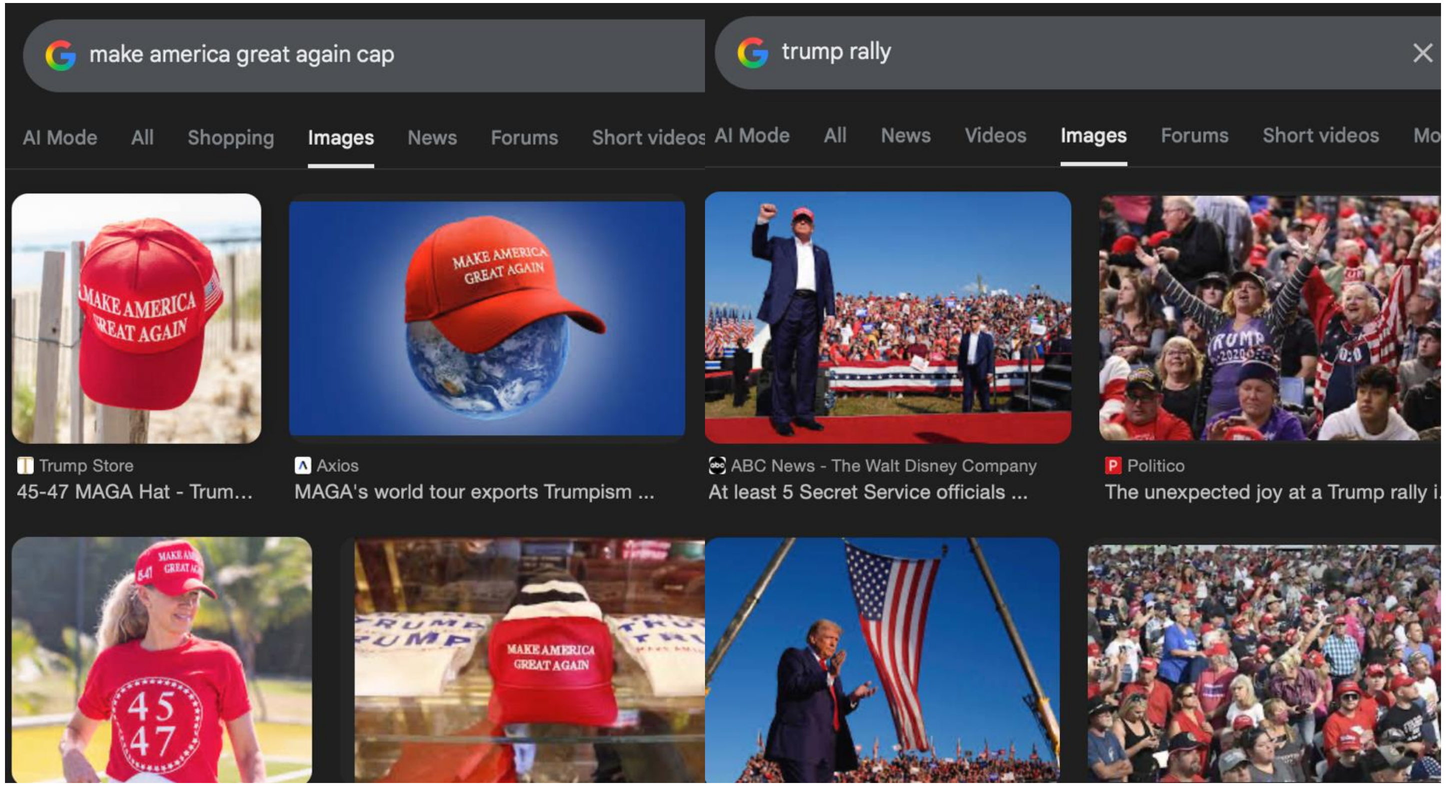
Challenges in political object detection




Creating a headwear classifier

Task: To predict three different political headwear: MAGA hats, Pussyhats, and Police headwear

1. **In-Distribution** Data Collection using **event** keywords



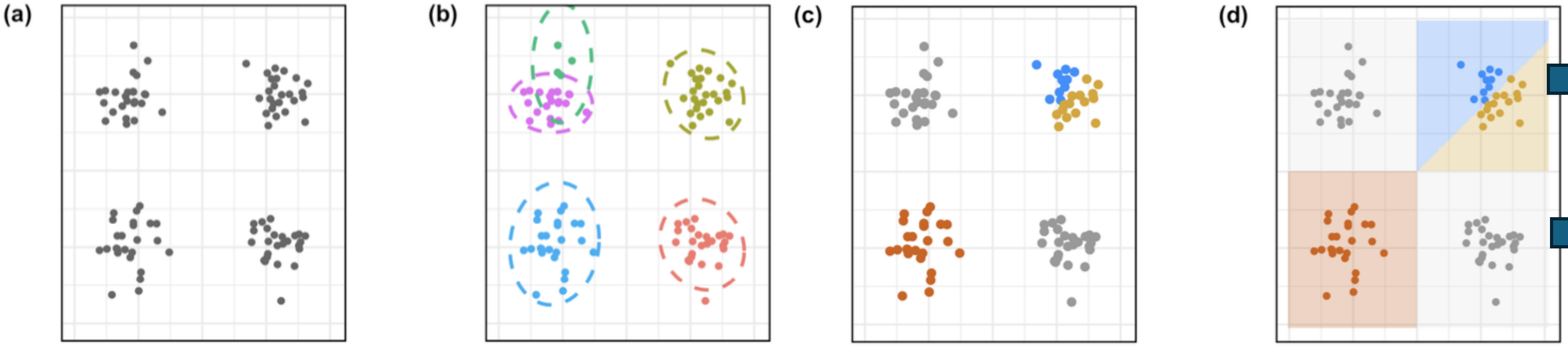
Keyword: {“MAGA hat”, “Pussyhat”, “Police headwear”}
Keyword: {“January 6 Capitol Attack”, “2017 Women’s March”, “Police Protester Confrontation”}

- In-Distribution Data Collection
- Open-Vocabulary Object Detection
- Image Segmentation
- Clustering
- Domain Knowledge-guided Labeling
- Training
- Weakly-Supervised Evaluation

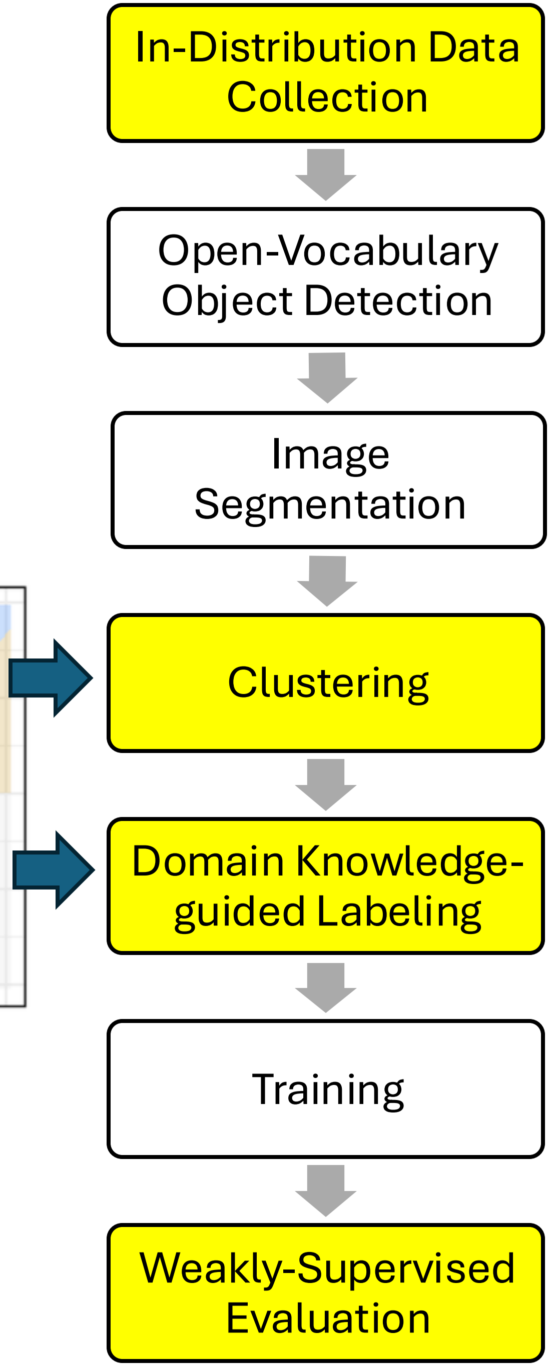
2. Open-Vocabulary Object Detection
3. Image Segmentation
4. Clustering

In-Distribution Data Collection
↓
Open-Vocabulary Object Detection
↓
Image Segmentation
↓
Clustering
↓
Domain Knowledge-guided Labeling
↓
Training
↓
Weakly-Supervised Evaluation

4. Clustering
5. Domain Knowledge-Guided Labeling



Raw segments | Clustering | Domain knowledge-guided labeling | Training

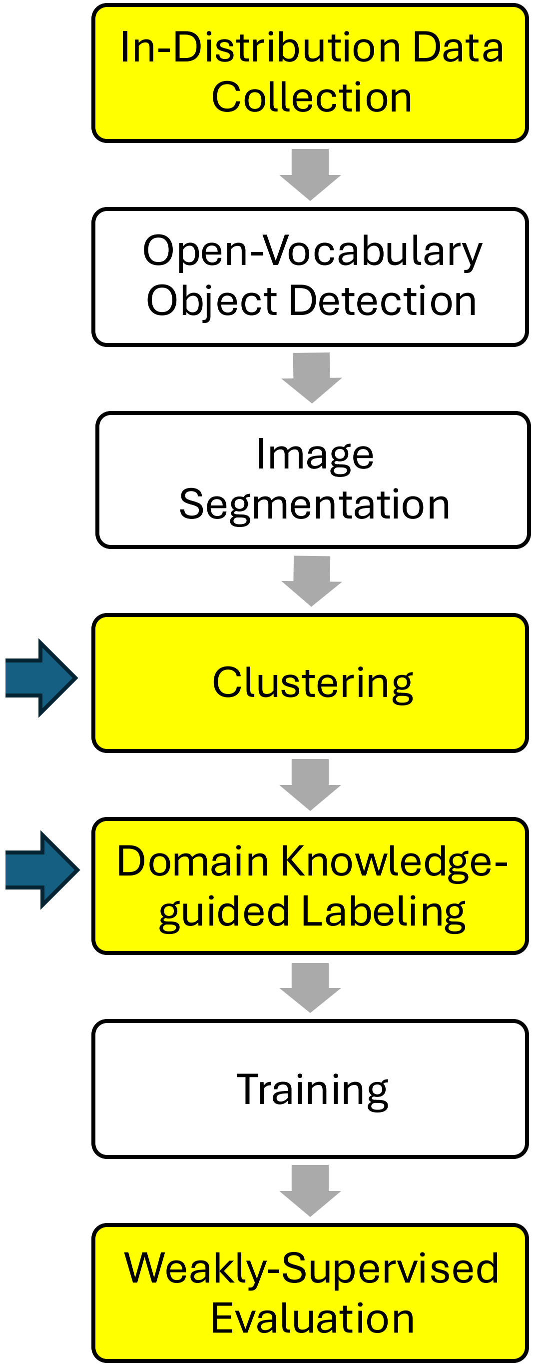


5. Domain Knowledge-Guided Labeling

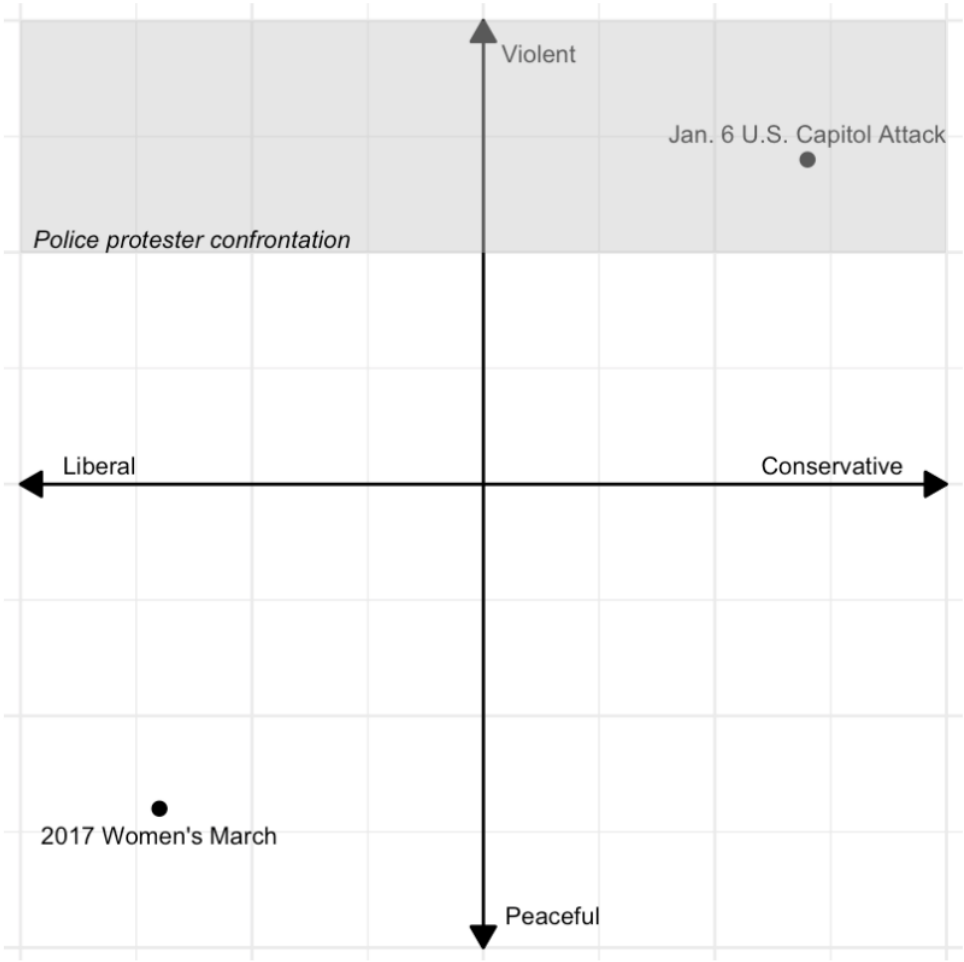
Algorithm 1 Labeling strategy for clustered samples

Require: Cluster assignments and originating query metadata
Ensure: Label for each sample

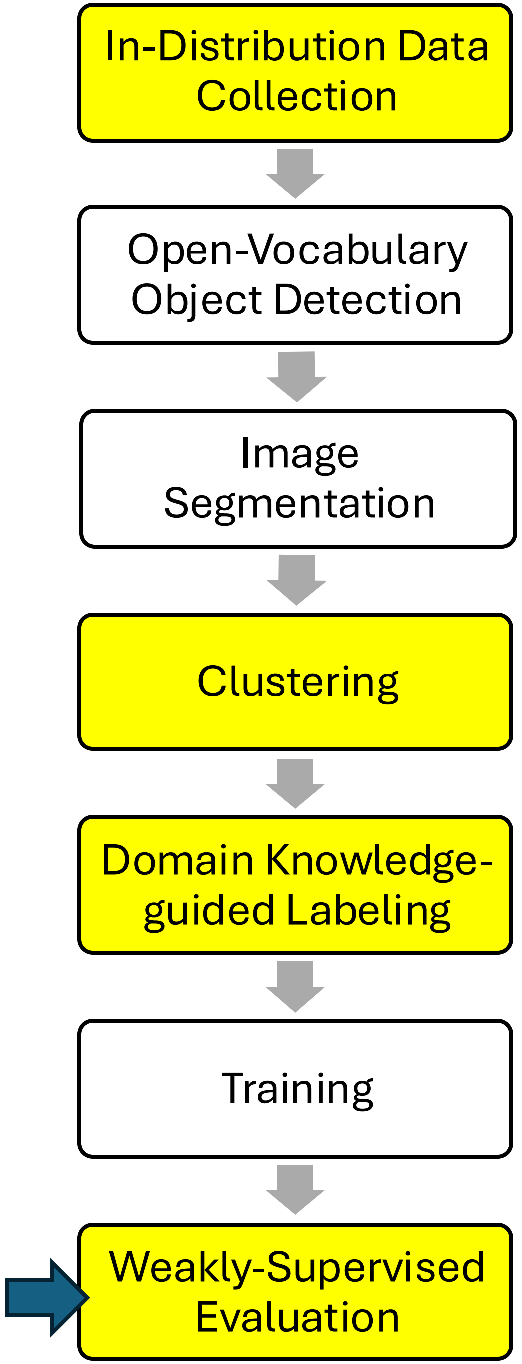
for all samples **do**
  if Cluster $\in \{-1, 2\}$ **then**
    Label $\leftarrow$ "Unknown"
  else if Cluster $= 1$ **then**
    Label $\leftarrow$ "Police"
  else if Cluster $= 0$ **then**
    if Query = '2017 Women's March' **then**
      Label $\leftarrow$ "Pussy"
    else if Query = 'January 6 Capitol Attack' **then**
      Label $\leftarrow$ "MAGA"
    end if
  end if
end for



7. Domain Knowledge-Guided Weakly-Supervised Evaluation



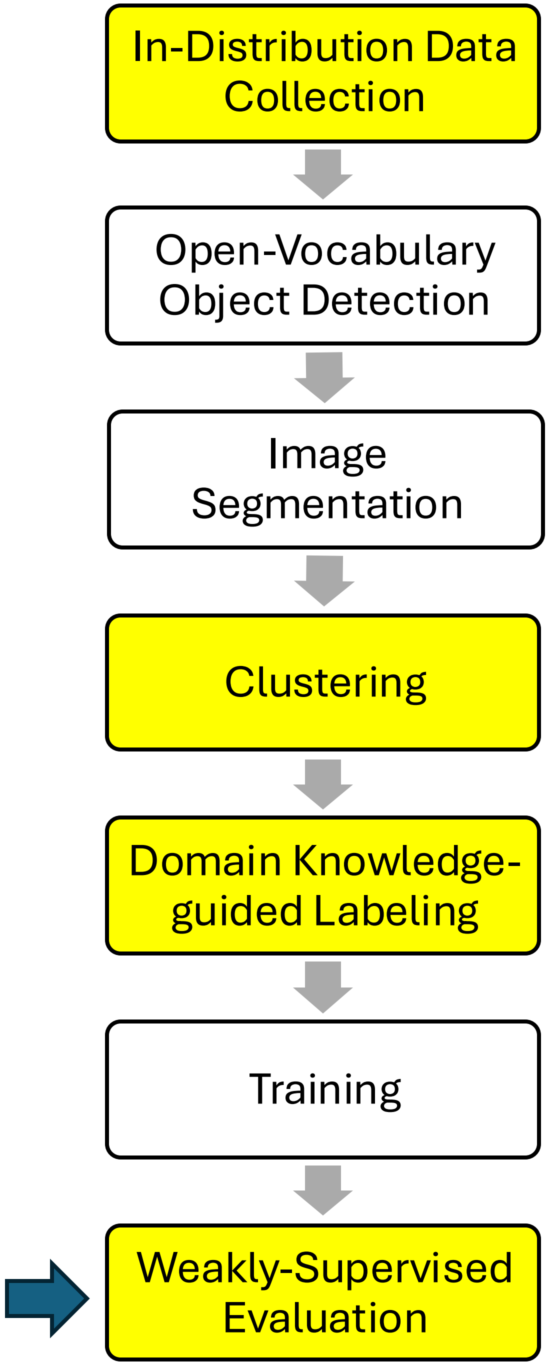
Domain knowledge

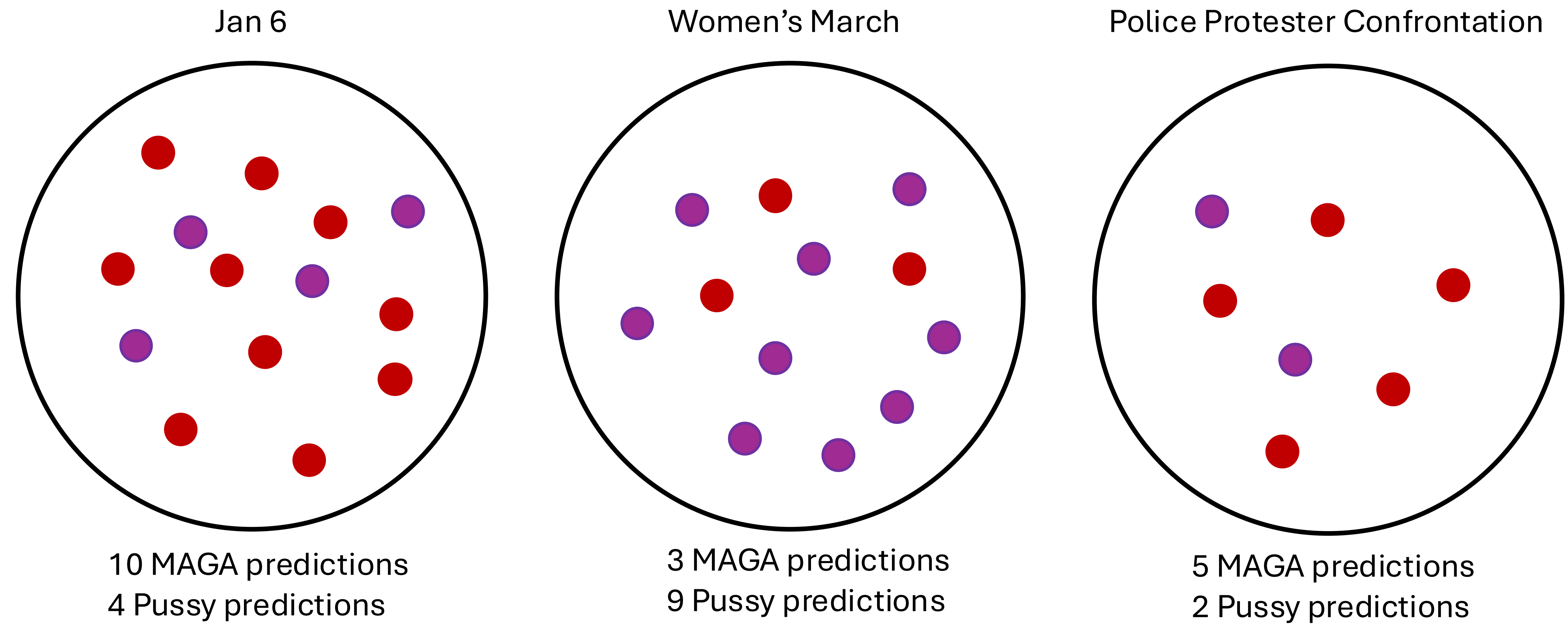


8. Domain Knowledge-Guided Weakly-Supervised Evaluation

- Weak supervised because:
  - GT labels are from metadata (Noise)
  - Not all predictions have GT labels (Sparsity)

- Metric: F1 score
  - $F1 = \frac{2 * Precision * Recall}{Precision + Recall}$
    - $Precision = \frac{TP}{TP+FP}$
    - $Recall = \frac{TP}{TP+FN}$

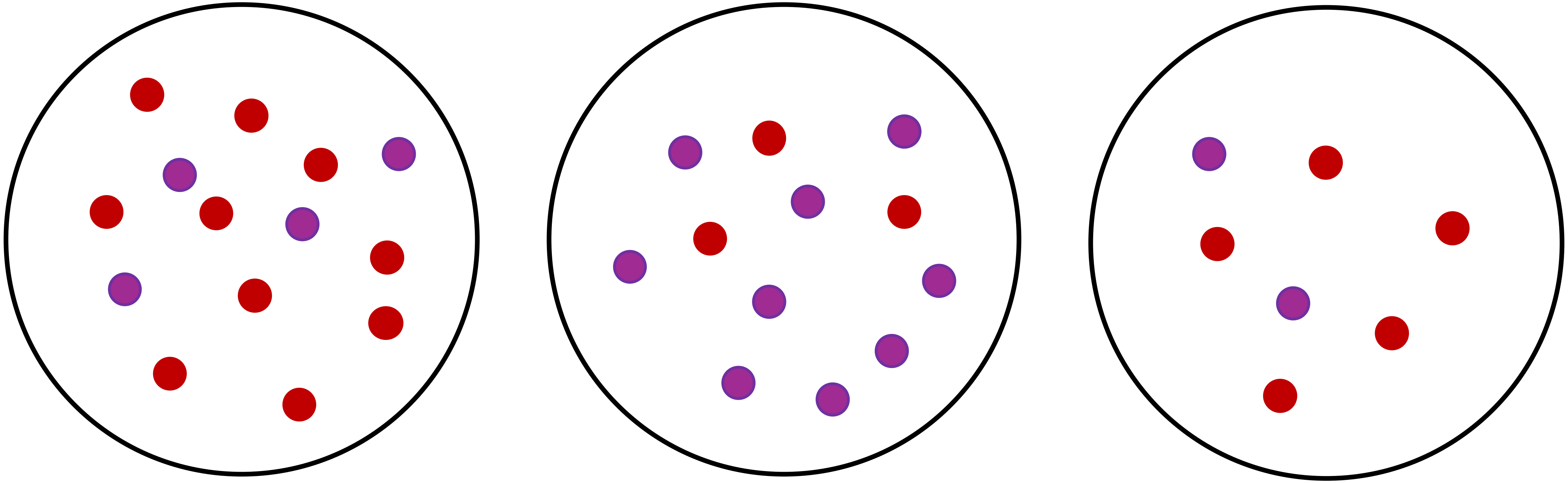




$$Precision_{MAGA} = \frac{TP}{TP + FP} = \frac{MAGA(Jan6)}{MAGA(Jan6) + MAGA(Women)} = \frac{10}{10+3} = \frac{10}{13}$$

$$Recall_{MAGA} = \frac{TP}{TP + FN} = \frac{MAGA(Jan6)}{MAGA(Jan6) + Pussy(Jan6)} = \frac{10}{10+4} = \frac{10}{14}$$

Jan 6 | Women's March | Police Protester Confrontation



10 MAGA predictions
4 Pussy predictions

3 MAGA predictions
9 Pussy predictions

5 MAGA predictions
2 Pussy predictions

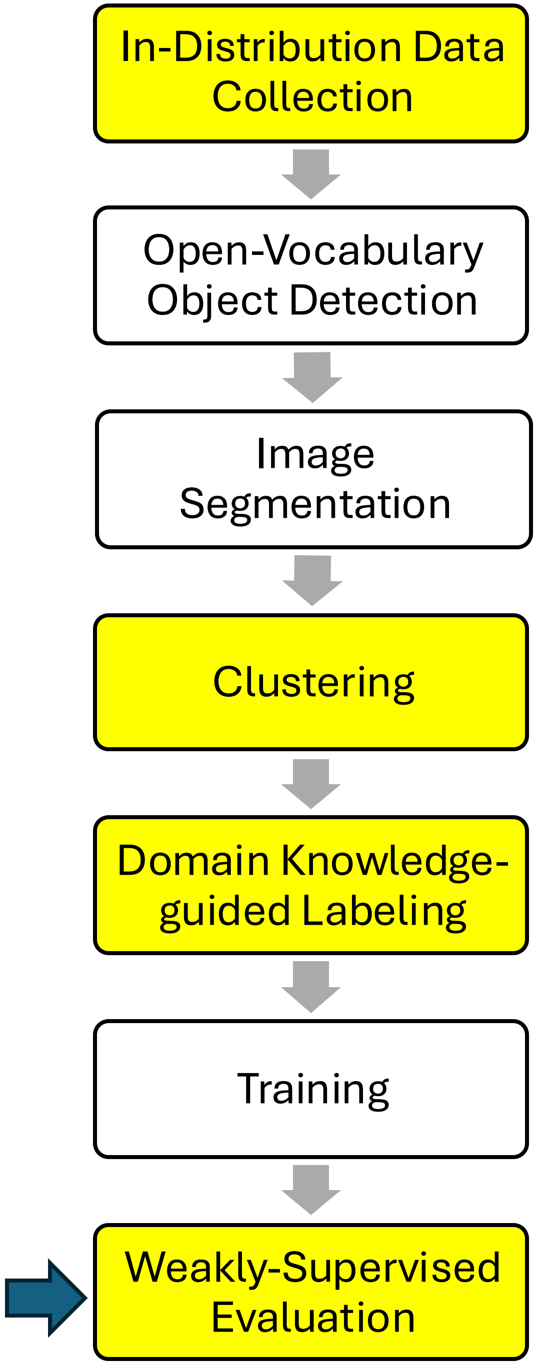
$$Precision_{Pussy} = \frac{TP}{TP + FP} = \frac{Pussy(Women)}{Pussy(Women) + Pussy(Jan6) + Pussy(Confrontation)} = \frac{9}{9 + 4 + 2} = \frac{9}{15}$$

$$Recall_{Pussy} = \frac{TP}{TP + FN} = \frac{Pussy(Women)}{Pussy(Women) + MAGA(Women)} = \frac{9}{9 + 4} = \frac{9}{12}$$

8. Domain Knowledge-Guided Weakly-Supervised Evaluation

Model	Within-cluster variance	F1
Proposed	1.90	0.968
Alternative 1: Query labels	3.32	0.882
Alternative 2: Cluster labels	1.69	N/A

Table 2: Quantitative results for three labeling approaches.



Application 1: *Objectscores*

- Data: 5k images from the 11 largest liberal and 9 largest conservative U.S. protests since 2017, identified by Crowd Counting Consortium U.S. Protest Event Data

Liberal (Labeled as -1)	Conservative (Labeled as 1)
March for Life	
Pro-Choice	Proud Boys
Women Lives Matter	Pro-Trump
Pro-Palestine	Back the Blue
Women's March	Pro-Life
Anti-Trump	Confederacy
Pride	MAGA
Black Lives Matter	Anti-Vax
Planned Parenthood	White Lives Matter
Drag Queen Story Hour	1776 Restorative Movement
Gaza	

Table 3: Keywords for Image Search

7 General Objects (w/o Subcat.)	Hat / Helmet (w/ 15 Subcat.)	Flag (w/ 5 Subcat.)	Signs / Banners (w/ 8 Subcat.)	Buildings (w/ 2 Subcat.)
man woman boy girl beard drag fire	Orange beanie Bright green hat Police hat Blue hat Beret Proud Boys hat Black beanie March for Life beanie Ski mask White bike helmet MAGA hat American beanie Trump hat Pussy hat Unknown	Rainbow American March for Lives Confederacy Unknown	Anti-abortion [Socialist Alternative Fight Racism] Box Pro-Trump Traffic sign [Don't Let McConnell Trump] Trans Rights Now Unknown	White House Unknown

Table 4: List of objects that can be detected by my customized political detection tool. Objects in square brackets are divided into two lines to fit the table.

Resulting data

Image ID	Keyword	Political Leaning (1 if Conservative; -1 if Liberal)	Political Object 1	Political Object 2	Political Object 3	Political Object 4	Political Object 5
1	Black Lives Matter	-1	3	4	0	2	1
2	Pro-Life	1	0	0			
3							
4							
5							
6							
7							

Application 1: *Objectscores*

1. $\text{Log Odds Score}(o) = \log\left(\frac{P(o|Conservative)}{P(o|Liberal)}\right)$
2. $Weighted\ Probability\ Score(o) = P(o|Conservative) * 1 + P(o|Liberal) * -1$

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Application 2: Objects as regression variables

- Data: 730 front images from WSJ and NYT on BLM protests from June 2020
  - 200 from WSJ; 530 from NYT
- Objects as **dependent** variable
  - How do outlets portray BLM protests differently?
    - $\boldsymbol{PolicePresence} = \alpha + \beta(Outlet\ ID = Protester\ ID) + \epsilon$
      - *PolicePresence = 1{police_hat + police_helmet + bike_helmet >0}*
- Objects as **independent** variable
  - How do the presence and magnitude of police officers, offensive and defensive gears in BLM images affect viewer violence perception?
    - $Perceived\ Violence = \alpha + \beta 1\boldsymbol{Police} + \beta 2\ \boldsymbol{Baton} + \beta 3\boldsymbol{Shield} + \epsilon$

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Limitations and Contributions

- *Objectscores* lack confidence intervals
  - Solution: bootstrapping
- Model prediction bias and variance are ignored
  - Solution: Maranca et al (2025)
- **Rare** political objects are not clustered
  - Solution: Data collection from videos

- Addresses the **gap** between current vision tools and the needs of political scientists
- Develops **domain knowledge-guided** strategies for:
  - Data annotation
  - Model evaluation
- Showcases potential usages with the two **most used** analyses:
  - Ideological scaling
  - Regression